

Coding & Solution

Date	05 July 2026
Team ID	xxxxxx
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	[Paste your GitHub / GitLab or hosted link here]
Programming Language(s)	Python
Framework(s) Used	Streamlit
Key Features Implemented	Voice upload, speech-to-text conversion, AI concept analysis, audio evaluation, score calculation, charts, PDF report generation
Pending / Incomplete Features	Future enhancements include multilingual support, real-time voice analysis, and advanced user progress tracking
Setup / Run Instructions	Install dependencies from requirements.txt and run application using streamlit run app.py

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The VBCUA application successfully implements an AI-powered voice analysis system using Streamlit and Python. The solution integrates speech recognition, NLP-based semantic evaluation, audio processing, and automated report generation to provide effective learning feedback.